Biochemistry Cume

Oct. 18, 2002

1. (10 points) The Michelis-Menten equation

$$v = \frac{V_{max}(S)}{K_m + (S)}$$

is the most widely used equation describing the kinetics of enzyme catalyzed reactions. It is based on the following kinetic scheme.

$$E + S \underset{k_{-1}}{\overset{k_1}{\leftrightarrows}} ES \xrightarrow{k_2} E + P$$

In terms of the rate constants in this kinetic scheme, how are v and Vmax defined?

2. (10 points) There are two commonly used ways to derive the Michaelis-Menten equation describing enzyme kinetics. Most textbooks utilize an approach involving a "steady-state" approximation. In class, we used a slightly different approach involving a "rapid-equilibrium" approximation. How do the basic assumptions of these two approaches differ?

3. (20 points) Using either of these approaches, derive the Michaelis Menten equation.

4. (20 points) Most biochemistry classes also describe analysis of competitive and noncompetitive inhibition. Describe the difference between these two types of inhibition based on the kinetic scheme in #1 and the terms in the Michaelis-Menten equation affected by each type of inhibitor

5. (20 points) There is a third type of enzyme inhibition called uncompetitive inhibition in which the inhibitor binds only to the ES complex. Defining the Ki as the dissociation constant for the reaction

$$ESI \leftrightarrows ES + I$$

derive the Michaelis-Menten-like equation for the case of an uncompetitve inhibitor and the corresponding Lineweaver-Burk transformation.

6. (20 points) Based on your results in #5, sketch a Lineweaver-Burk plot showing how it could be used to diagnose a situation involving au uncompetitive inhibitor (how would such a plot for an uncompetitve inhibitor differ from those for competitive and noncompetitive inhibitors?)